Existentially Consistent Skolemized Reasoning for Complex Query Answering on Knowledge Graphs

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Abstract

Complex Query Answering (CQA) over incomplete Knowledge Graphs formalizes compositional reasoning as Existential First-Order logic with one free variable (EFO_1), where existential variables link multiple atomic constraints within a query. Maintaining existential consistency helps prevent reasoning shortcuts that satisfy only part of these constraints. We identify a Grounding–Skolemization dichotomy in existing CQA approaches. Grounding-based methods preserve logical equivalence but incur combinatorial enumeration costs, whereas Skolemization-based methods reason efficiently in latent space but risk existential inconsistency due to query-level representation entanglement. To address this tradeoff, we propose existentially consistent Skolemized reasoning, instantiated by the Logic-constrained Vector Symbolic Architecture (LVSA). LVSA uses vector-symbolic operations to compose atomic constraints and differentiable Skolemization to construct and reuse a constraint-conditioned latent state for each existential variable. Logical negation is handled through a logic-regularized neural negator. Theoretically, LVSA is structurally expressive for EFO_1 queries with acyclic query graphs, with favorable complexity. Experiments show that LVSA outperforms prior Skolemization-based methods in reasoning performance and empirical existential consistency while achieving substantially lower inference costs than Grounding-based approaches.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**.

Keywords

Complex Query Answering, Knowledge Graphs, Neuro-Symbolic AI, First-Order Logic.

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1 Introduction

Knowledge Graphs (KGs) organize factual knowledge as graphs of entities and relations [8, 28], supporting a wide range of knowledge-intensive applications [7, 34]. However, real-world KGs are inherently incomplete: many true facts are missing, making simple triple retrieval insufficient for answering compositional questions [14]. Complex Query Answering (CQA) addresses this challenge by answering logical queries over incomplete KGs. These queries are commonly formalized in Existential First-Order logic with one free variable (EFO_1), which supports conjunction, disjunction, negation, and existential quantification. In the era of Large Language Models (LLMs), CQA can serve as a crucial complement to LLMs for applications requiring efficient, logically grounded, and controllable reasoning over structured knowledge [12, 13].

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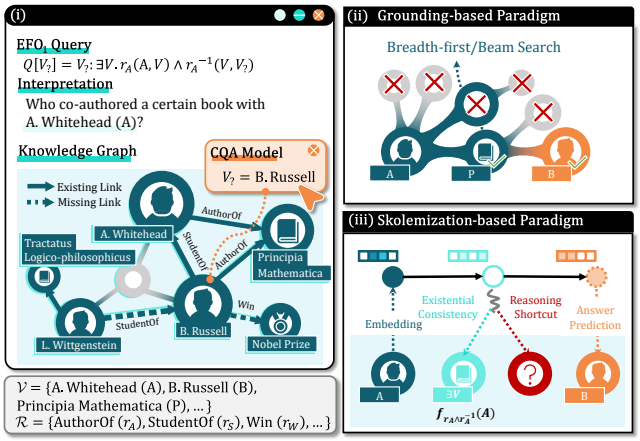


Figure 1: (i) An example KG and an EFO_1 query. -1 denotes the inverse relation. (ii) Grounding-based reasoning workflow. (iii) Skolemization-based reasoning workflow.

In EFO_1, existential variables link multiple atomic constraints within a query. Maintaining a shared assignment for each such variable across these constraints is therefore essential for logical fidelity. For instance, in the two-hop query shown in Figure 1(i), the existential variable V must simultaneously satisfy two atomic constraints: $r_A(A, V)$ and $r_A^{-1}(V, V_?)$. Re-examining existing CQA methods through the lens of first-order logic [32], we identify two fundamentally different strategies for handling existential variables, which we characterize as a *Grounding–Skolemization dichotomy*. This dichotomy reveals a central tradeoff between logical fidelity and inference efficiency in existing CQA systems.

One line of work decomposes complex queries into atomic subqueries, enumerates candidate entities for each existential variable, and aggregates the predictions of grounded atomic queries via fuzzy logic [1, 3, 15, 40]. This strategy corresponds to *Grounding* in classical logic, where existential variables are eliminated by substituting all candidate entities and evaluating the resulting propositional formula. We refer to these as *Grounding-based* methods (Figure 1(ii)). By instantiating existential variables, grounding preserves logical equivalence and existential consistency. However, its computational complexity grows exponentially with the number of dependent existential variables, rendering such methods impractical for complex query structures over large KGs [23]. Moreover, exhaustive enumeration limits generalization in knowledge-sparse settings [10].

Another line of work performs compositional reasoning directly in a continuous embedding space [11, 25, 33, 37]. From the perspective of first-order logic, these methods can be viewed as implicitly approximating *Skolem functions*, which construct latent states intended to encode candidate assignments for existential variables without explicit enumeration. We therefore refer to them as *Skolemization-based* methods (Figure 1(iii)). Compared with grounding, Skolemization-based methods scale efficiently by avoiding combinatorial entity enumeration. However, existing methods typically construct query-centric embeddings for answer prediction through geometric operators or deep neural architectures. Under this design, latent states of existential variables become progressively entangled within evolving query representations, making it difficult to preserve existential consistency across reasoning steps. Consequently, the model may take reasoning shortcuts in the latent space, such as partially satisfying logical constraints while still producing plausible answers [17].

To address these limitations, we propose *existentially consistent Skolemized reasoning*, instantiated through the **L**ogic-constrained **V**ector **S**ymbolic **A**rchitecture (**LVSA**)[1]. Instead of collapsing all constraints into a query-centric representation, LVSA organizes reasoning around persistent per-variable latent states, thereby preserving existential consistency at the structural level. To realize this principle, LVSA builds on Vector Symbolic Architectures (VSA) [27], using algebraic operations to represent and compose atomic relational constraints in the latent space. Built on this structured compositional framework, a differentiable Skolemization module constructs a constraint-conditioned latent state for each existential variable. During compositional reasoning, LVSA persistently reuses the same state at every subsequent occurrence of that variable and sequentially incorporates its relevant atomic constraints. Furthermore, since modeling logical negation in latent space is challenging, a neural negator with logic-guided regularization handles hard negation by constraining logical behaviors such as the double-negation law. Together, these neuro-symbolic components preserve the efficiency advantages of latent reasoning while preventing the shortcuts that undermine logical fidelity.

Theoretical analysis establishes LVSA's ability to construct and maintain persistent existential-variable states, with favorable computational complexity compared with Grounding-based methods. Experimental results show that LVSA consistently outperforms state-of-the-art Skolemization-based baselines in reasoning accuracy, while achieving higher throughput, stronger generalization under sparse knowledge conditions, and substantially improved empirical existential consistency throughout reasoning steps.

2 Preliminary

2.1 Problem Formalization

A **Knowledge Graph (KG)** can be denoted as $\mathcal{G} = \{\mathcal{V}, \mathcal{R}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{R}$ are the sets of all entities and relations, respectively, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ is a set of true assertions in the form of triples. Under the open-world assumption [26], an assertion triple of $(e_h, r, e_t) \notin \mathcal{E}$ with $e_h, e_t \in \mathcal{V}$ and $r \in \mathcal{R}$ may also be true.

Complex Queries on KGs can be formalized as **Existential First-Order predicate formulas with one free variable (EFO_1)**. Formally, a valid EFO_1 query Q consisting of an anchor entity set $\mathcal{V}_h \subseteq \mathcal{V}$, a set of existentially quantified variables $\mathcal{V}_\exists = \{V_1, V_2, \ldots, V_k\}$, and a free variable $V_?$, can be defined in the Disjunctive Normal Form (DNF) as follows:

$$Q[V_?] = V_? : \exists V_1, \ldots, V_k : c_1 \vee c_2 \vee \cdots \vee c_n, \tag{1}$$

where each c is a conjunctive query, that is, $c_i = q_{i1} \wedge q_{i2} \wedge \cdots \wedge q_{im}$. And each literal q is an atomic formula or its negation, that is, $q_{ij} = r(V', V)$ or $q_{ij} = \neg r(V', V)$, with head $V' \in \mathcal{V}_h \cup \mathcal{V}_\exists$, tail $V \in \{V_?\} \cup \mathcal{V}_\exists$, and relation $r \in \mathcal{R}$. The answer set of the complex query $Q[V_?]$ is denoted as $A[Q] \subseteq \mathcal{V}$.

[1]The source code of LVSA is available at https://github.com/luyy9apples/LVSA.

This work focuses on EFO_1 queries whose variable-dependency graphs are directed acyclic graphs (DAGs), referred to as DAG-structured EFO_1 queries. This query class forms the basis of widely used CQA benchmarks [11, 24, 25].

2.2 EFO_1 Reasoning

A fundamental challenge in EFO_1 reasoning lies in handling existentially quantified variables (hereafter, *existential variables*). Grounding and Skolemization are two classical transformations relevant to this problem, respectively associated with Herbrand's Theorem and Skolem's Theorem [32].

Definition 1 (Grounding). *Given an EFO$_1$ formula $\exists V : r(V', V)$, grounding converts it to a ground formula by enumerating all candidate entities for substitutions $\{e/V | e \in \mathcal{V}\}$:*

$$\exists V : r(V', V) \overset{G}{\Rightarrow} \bigvee_{e \in \mathcal{V}} r(V', e). \tag{2}$$

Property 1 (Properties of Grounding). *Let F be an EFO$_1$ formula and $F \overset{G}{\Rightarrow} F_G$:*

a. ***Equivalence****: F and F_G are logically equivalent.*

b. ***Complexity****: Let F involve k existential variables.*

(i) ***Best-case****: When existential variables are independent, grounding performs k separate substitutions, yielding linear complexity $O(k \cdot |\mathcal{V}|)$.*

(ii) ***Worst-case****: When existential variables form a chain of dependencies, grounding requires chained substitutions, yielding exponential complexity $O(|\mathcal{V}|^k)$.*

Definition 2 (Skolemization). *Given EFO$_1$ formulas $F_D = \exists V_D : r(V', V_D)$ and $F_I = \exists V_I : r(V_I, V)$ with dependent existential variable V_D and independent existential variable V_I, Skolemization eliminates the existential quantifier by respectively introducing a Skolem function $f_r(V')$ and a Skolem constant $f_r()$, as follows:*

$$F_D \overset{S}{\Rightarrow} r\left(V', f_r(V')\right), \; F_I \overset{S}{\Rightarrow} r\left(f_r(), V\right). \tag{3}$$

Property 2 (Properties of Skolemization). *Let F be an EFO$_1$ formula and $F \overset{S}{\Rightarrow} F_S$:*

a. ***Satisfiability****: F is satisfiable $\Leftrightarrow$ F_S is satisfiable. $F_S \models F$ but the converse ($F \models F_S$) is not true in general.*

b. ***Complexity****: When F involves k existential variables, the complexity of Skolemization is $O(k)$.*

Example 1. *Consider the complex query $Q[V_?] = V_? : \exists V : r_A(A, V) \wedge r_A^{-1}(V, V_?)$ illustrated in Figure 1. Grounding-based reasoning is defined by*

$$Q[V_?] \overset{G}{\Rightarrow} Q_G[V_?] = V_? : \Big(r_A(A, P/V) \wedge r_A^{-1}(P/V, V_?)\Big) \vee \Big(r_A(A, B/V) \wedge r_A^{-1}(B/V, V_?)\Big) \vee \cdots$$

In contrast, classical Skolemization yields

$$Q[V_?] \overset{S}{\Rightarrow} Q_S[V_?] = V_? : r_A(A, f_{r_A}(A)) \wedge r_A^{-1}(f_{r_A}(A), V_?).$$

As we shall discuss in Section 4, LVSA implements a latent analogue of this construction by conditioning the state of an existential variable on its relevant atomic constraints, yielding $f_{r_A \wedge r_A^{-1}}(A)$. This conditioning provides a structural mechanism for reusing the same variable state across the constraints, rather than a guarantee that the decoded state is a model-theoretically valid witness.

3 The Grounding-Skolemization Dichotomy

We characterize existing CQA methods by how they handle existential variables, leading to the Grounding-Skolemization dichotomy under a common operator framework $(\varphi, \{g_P, g_\wedge, g_\vee, g_\neg, g_\exists\}, \phi_E)$. Here, φ is the embedding function, $\{g_P, g_\wedge, g_\vee, g_\neg, g_\exists\}$ denote relational projection, conjunction, disjunction, negation, and existential quantification, respectively, and ϕ_E is the similarity measure.

Grounding-based Methods [1, 3, 9, 15, 18, 36, 40] ground EFO_1 queries into propositional formulas by explicitly instantiating existential variables via entity enumeration. Their core components consist of a link predictor $\phi : \mathcal{V} \times \mathcal{R} \times \mathcal{V} \mapsto [0, 1]$, which serves as both the relational projection operator g_P and the final scorer ϕ_E, together with fuzzy logical operators $\{\top : g_\wedge, \; \bot : g_\vee, \; \text{Neg} : g_\neg\}$ [20]. Operationally, $g_\exists$ aggregates scores over explicit entity assignments, either exhaustively [3] or over a pruned candidate set [1]. For Example 1 with $V_? = \text{B}$, the answer score is computed by aggregating over all grounded substitutions of the existential variable V:

$$E\left(Q_G[\text{B}]\right) = \Big(\phi(\text{A}, r_A, \text{P}) \top \phi(\text{P}, r_A^{-1}, \text{B})\Big) \bot \Big(\phi(\text{A}, r_A, \text{B}) \top \phi(\text{B}, r_A^{-1}, \text{B})\Big) \bot \cdots$$

By Property 1a, exhaustive grounding preserves logical equivalence and uses the same entity assignment for every occurrence of an existential variable. However, by Property 1b, its computational complexity grows exponentially with the number of dependent existential variables, rendering exhaustive grounding impractical for large-scale CQA. Pruned approximations improve efficiency by sacrificing completeness over the space of entity assignments. Moreover, in knowledge-sparse settings, exhaustive grounding may amplify noisy predictions in reasoning chains.

Skolemization-based Reasoning Methods [2, 11, 25, 33, 37–39, 41] avoid explicit enumeration by performing compositional reasoning directly in a continuous latent space. From this perspective, these methods approximate Skolem functions by constructing latent representations of candidate assignments for existential variables. Formally, they define transformation operators $\{g_P, g_\wedge, g_\vee, g_\neg, g_\exists\}$ for relational projection, logical composition, and latent existential reasoning. For Example 1, the query embedding and answer score are computed as:

$$\varphi(Q) = g_P\left(f_{r_A}(\text{A}) \mid r_A^{-1}\right), \text{where } f_{r_A}(\text{A}) \approx g_\exists(\text{A} \mid Q),$$
$$E\left(Q_S[\text{B}]\right) = \phi_E\left(\varphi(Q), \varphi(\text{B})\right).$$

By avoiding enumeration over joint entity assignments, these methods eliminate the direct $O(|\mathcal{V}|^k)$ dependence of exhaustive grounding. Geometric methods represent candidate sets as boxes [24], beta distributions [25], cones [38], or fuzzy sets [6, 16], and progressively compose them into query-level representations. Neural methods instead construct latent representations through optimization or learned architectures: CQD-CO [1] directly optimizes existential embeddings, LMPNN [33] and CLMPT [37] use GNN-based message passing, and Transformer-based methods fuse query information through attention [39, 41]. Despite these architectural

differences, prior methods have not explicitly addressed existential consistency throughout compositional reasoning. Consequently, the coherence of intermediate existential assignments remains difficult to inspect, and models may produce plausible answers through reasoning shortcuts.

4 Methodology

To overcome the tradeoff between logical fidelity and inference efficiency revealed by the Grounding–Skolemization dichotomy, we propose the **L**ogic-constrained **V**ector **S**ymbolic **A**rchitecture (**LVSA**). Unlike prior Skolemization-based methods that rely on evolving query-level representations, LVSA adopts a variable-centric reasoning paradigm. Its conceptual contribution lies in explicitly separating persistent existential states from query-level composition, while VSA operations and neural modules provide the corresponding implementation.

Given a DAG-structured EFO_1 query, LVSA follows a topological reasoning order so that the predecessor dependencies of a variable are resolved before its latent state is constructed. Once constructed, the state of each existential variable is maintained and reused throughout subsequent reasoning steps. This design preserves variable identity at the structural level and avoids inconsistent state updates across occurrences of the same variable.

To realize this reasoning process, LVSA separates logical composition from existential state construction. Vector symbolic operations algebraically represent and compose atomic relational and logical constraints (Section 4.1), while differentiable Skolemization constructs latent states intended to encode candidate witnesses under the relevant constraints (Section 4.2). A logic-regularized neural negator handles logical negation in the latent space (Section 4.3), and the entire framework is trained jointly under a curriculum learning strategy (Section 4.4)

4.1 Vector Symbolic Query Encoding

Embedding Function (φ): Building on the success of complex-valued vectors in both Knowledge Graph Embedding (KGE) models [19, 29, 31] and VSA [21, 22], we map entities into a complex vector space. For an entity $e \in \mathcal{V}$, its complex-valued embedding is defined by

$$\varphi(e) \triangleq \text{Re}(e) + \text{Im}(e)i, \tag{4}$$

where the trainable real component $\text{Re}(e) \in \mathbb{R}^d$ and imaginary component $\text{Im}(e) \in \mathbb{R}^d$ replace traditional VSA's random vectors to capture semantic correlation among entities. We also define the concatenated vector representation of $\varphi(e)$ as $v_\varphi(e) \triangleq [\text{Re}(e); \text{Im}(e)]$.

Relational Projection (g_P): Each relation $r \in \mathcal{R}$ is represented in the same complex embedding space as entities, i.e., $\varphi(r) \triangleq \text{Re}(r) + \text{Im}(r)i$. Relational projection is then realized by the **Binding ($\otimes$)** operator, implemented as the Hadamard product ($\odot$) between the relation and head representations:

$$\begin{aligned}\varphi(r \otimes V') = &\left[\text{Re}(r) \odot \text{Re}(V') - \text{Im}(r) \odot \text{Im}(V')\right] \\ &+ \left[\text{Re}(r) \odot \text{Im}(V') + \text{Im}(r) \odot \text{Re}(V')\right] i.\end{aligned} \tag{5}$$

Intuitively, binding emphasizes specific attributes of each entity under different relations. For instance, StudentOf $\otimes$ A $\approx$ B and AuthorOf $\otimes$ A $\approx$ P distinguish A. Whitehead's roles as an advisor versus an author.

Logical Conjunction ($g_\wedge$): Inspired by VSA, we model $g_\wedge$ through **Bundling (+)**. Bundling combines multiple constraint representations while retaining information from individual inputs, and naturally preserves the commutativity and associativity of conjunctive composition. Given a conjunctive query $Q_\wedge = r_1(V_1', V) \wedge \cdots \wedge r_N(V_N', V)$, we first compute the bundled representation:

$$\tilde{\varphi}(Q_\wedge) = \frac{1}{N} \sum_{j=1}^{N} \varphi(r_j \otimes V_j'). \tag{6}$$

Since repeated bundling may alter embedding magnitudes, we further normalize the bundled representation (NormAdd) to maintain scale consistency:

$$\varphi(Q_\wedge) = \text{NormAdd}\left[\varphi(r_1 \otimes V_1'), \ldots, \varphi(r_N \otimes V_N')\right] = \frac{L}{\tilde{L}} \cdot \tilde{\varphi}(Q_\wedge), \tag{7}$$

where $\tilde{L}$ is the average magnitude of the bundled representation $\tilde{\varphi}(Q_\wedge)$, and L is the average magnitude of the input embeddings.

Logical Disjunction ($g_\vee$): DNF can decompose the answer set of a disjunctive query into a union of the answer sets of its conjunctive sub-queries, i.e., $A[Q] = A[c_1] \cup \cdots \cup A[c_n]$ for $Q[V_?]$.

Similarity Measure (ϕ_E): We adopt the real part of the Hermitian inner product as the similarity between complex-valued vectors. Formally, the predicted score of a candidate entity $e \in \mathcal{V}$ for the complex query $Q[V_?]$ is defined as

$$\begin{aligned}E(Q_S[e]) &= \text{Re}\left(\langle \varphi(Q), \overline{\varphi(e)} \rangle\right) \\ &= \text{Re}\,(Q) \cdot \text{Re}\,(e) + \text{Im}\,(Q) \cdot \text{Im}\,(e)\,,\end{aligned} \tag{8}$$

where $\overline{\varphi(e)} \triangleq \text{Re}(e) - \text{Im}(e)i$ is the conjugate of $\varphi(e)$.

4.2 Differentiable Skolemization

LVSA realizes the existential operator $g_\exists$ with a differentiable Skolemization module that constructs a persistent latent state for each existential variable, conditioned on its relevant constraints. The state is reused throughout subsequent reasoning, allowing all constraints involving the same variable to operate on a shared representation. This variable-sharing mechanism parallels classical Skolemization, where all occurrences of an existential variable share the same Skolem term.

Independent Existential Variables are ones whose latent representations can be constructed directly from relation semantics, without relying on other variable states, as in

$$\exists V_I : r(V_I, V) \overset{S}{\Rightarrow} r\,(f_r(), V)\,.$$

LVSA therefore initializes the persistent latent state of V_I through a relation-aware neural network MLP_I:

$$\varphi(V_I) = \text{MLP}_\text{I}\left(v_\varphi(r)\right). \tag{9}$$

Dependent Existential Variables are ones whose latent states are conditioned on multiple logical constraints and previously constructed variable states, as in

$$F = \exists V_D : r_1(V', V_D) \wedge r_2(V_D, V),$$

where a valid witness for V_D must simultaneously satisfy both relational constraints. To incorporate both upstream and downstream

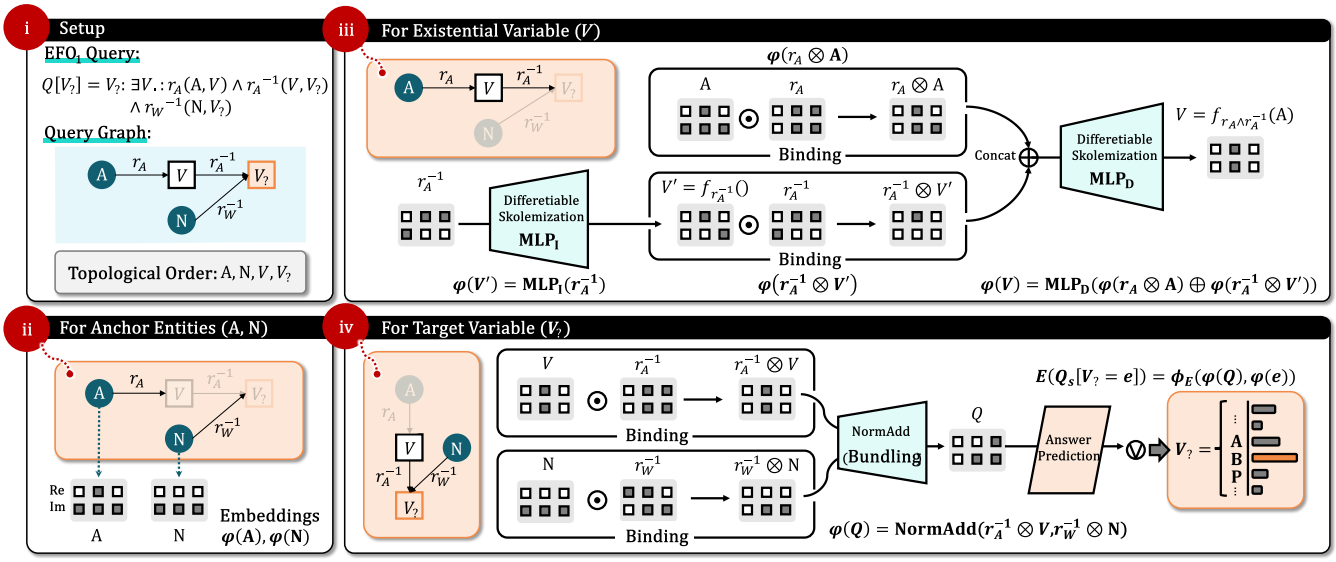


Figure 2: LVSA's inference process for a sample complex query $Q[V_?] = V_? : \exists V : r_A(\mathbf{A}, V) \wedge r_A^{-1}(V, V_?) \wedge r_W^{-1}(\mathbf{N}, V_?)$. Panels (i)–(iv) show the query graph, anchor embeddings, existential variable inference, and target variable inference, respectively.

constraints when constructing the latent state of V_D, LVSA represents them in a common form directed toward V_D. Specifically, the downstream constraint $r_2(V_D, V)$ is re-expressed as $r_2^{-1}(V, V_D)$, and a relation-aware auxiliary state V_I encodes the unavailable downstream context, yielding the following input canonicalization:

$$F \Rightarrow_* F^* = \exists V_D, V_I : r_1(V', V_D) \wedge r_2^{-1}(V_I, V_D). \tag{10}$$

Under this canonicalization, LVSA represents the shared state of V_D with the constraint-conditioned latent term $f_{r_1 \wedge r_2^{-1}}(V')$:

$$F^* \overset{S}{\Rightarrow} F_S^* = r_1\left(V', f_{r_1 \wedge r_2^{-1}}(V')\right) \wedge r_2^{-1}\left(f_{r_2^{-1}}(), f_{r_1 \wedge r_2^{-1}}(V')\right). \tag{11}$$

LVSA then constructs the latent state of V_D by jointly composing all relevant constraint representations:

$$\varphi(V_D) = \mathrm{MLP}_\mathrm{D}\left(v_\varphi(r_1 \otimes V') \oplus v_\varphi(r_2^{-1} \otimes V_I)\right). \tag{12}$$

where $\oplus$ denotes embedding concatenation.

By constructing a constraint-conditioned latent state for each existential variable, LVSA enforces structural existential consistency during latent reasoning. Combined with the algebraic compositionality of VSA, the proposed framework can structurally represent DAG-structured EFO_1 queries with favorable computational complexity (Section 5).

Example 2. *Figure 2 illustrates the sequential reasoning workflow of LVSA on the sample query.*

4.3 Logic-Regularized Neural Negation

Modeling logical negation in latent space is fundamentally challenging because semantic inversion is not naturally well-defined geometrically. For example, negating "cat" does not correspond to a distant concept such as "iris", but rather to a semantically related yet logically incompatible concept such as "dog" [15]. Consequently, logical negation cannot be reliably modeled through simple algebraic inversion operators in the embedding space.

To address this issue, LVSA employs a neural negator MLP_N that learns latent semantic inversion conditioned on the conjunctive context [25]. For a query involving logical negation $Q_\neg = Q_\wedge \wedge \neg q = Q_\wedge \wedge \neg r_{N+1}(V'_{N+1}, V)$, the embedding of $Q_\neg$ is computed by

$$\begin{aligned} \varphi(\neg(r_{N+1} \otimes V'_{N+1})) &= \mathrm{MLP}_\mathrm{N}\left(v_\varphi(Q_\wedge) \oplus v_\varphi(r_{N+1} \otimes V'_{N+1})\right), \\ \varphi(Q_\neg) &= \mathrm{NormAdd}\left[\varphi(Q_\wedge), \varphi(\neg(r_{N+1} \otimes V'_{N+1}))\right]. \end{aligned} \tag{13}$$

However, an unconstrained neural negator may learn degenerate shortcuts that violate fundamental logical behavior. LVSA therefore regularizes the neural negator using both satisfiability constraints and logical axiom regularization.

Satisfiability Constraint. If an entity t satisfies the negated query $Q_\neg$, the negated predicate $\neg q[t]$ should receive a high truth score. This constraint is enforced via the following satisfiability objective:

$$\mathcal{L}_\mathrm{NS}(\mathcal{B}_\neg; \alpha) = -\frac{\alpha}{|\mathcal{B}_\neg|} \sum_{Q_\neg \in \mathcal{B}_\neg} \ln \sigma\left(E(\neg q[t])\right), \tag{14}$$

where $\sigma(\cdot)$ denotes the sigmoid function and α controls the contribution of the satisfiability regularization term.

Logical Axiom Regularization. The neural negator is further regularized to approximately preserve two fundamental logical properties: the double negation law ($\neg\neg x \equiv x$) and the contradiction law ($x \wedge \neg x \equiv$ False). We interpret the zero vector $\mathbf{0}$ as a degenerate latent state corresponding to the absence of coherent satisfiable semantics. Accordingly, the logical regularization objective is defined as

$$\begin{aligned} \mathcal{L}_\mathrm{NL}(\mathcal{B}_\neg; \beta) = \frac{\beta}{|\mathcal{B}_\neg|} \sum_{Q_\neg \in \mathcal{B}_\neg} &\mathrm{MSELoss}\left(\varphi(\neg\neg q), \varphi(q)\right) \\ &+\mathrm{MSELoss}\left(\varphi(\neg q \wedge q), \mathbf{0}\right), \end{aligned} \tag{15}$$

where $\text{MSELoss}(x, y) = \|x - y\|_2^2$, and β controls the strength of logical regularization.

4.4 Training

The primary training objective of LVSA is a cross-entropy loss that encourages correct answer entities to obtain higher scores than negative samples. Given a training batch $\mathcal{B} = \{(Q, a)\}$, the objective is defined as

$$\begin{aligned}\mathcal{L}_{\text{CE}}(\mathcal{B}) &= \frac{1}{|\mathcal{B}|} \sum_{(Q,a)\in\mathcal{B}} H(Q, \mathcal{V}) \\ &= \frac{1}{|\mathcal{B}|} \sum_{(Q,a)\in\mathcal{B}} [-E\left(Q_S[a]\right) + \ln \sum_{e\in\mathcal{V}} \exp\left(E\left(Q_S[e]\right)\right)].\end{aligned} \tag{16}$$

To stabilize optimization and reduce overfitting, LVSA adopts a three-stage curriculum learning strategy [4] that progressively introduces increasingly complex logical operators:

(i) Train entity and relation embeddings ($\{\varphi(e)|e \in \mathcal{V}\} \cup \{\varphi(r)|r \in \mathcal{R}\}$) on atomic queries (***1p***): $\mathcal{L}_1 = \mathcal{L}_{\text{CE}}(\mathcal{B}_{\boldsymbol{1p}})$.
(ii) Fix pretrained embeddings and fine-tune the differentiable Skolemization module (MLP_I and MLP_D) on multi-hop queries (***2p*** and ***3p***): $\mathcal{L}_2 = \mathcal{L}_{\text{CE}}(\mathcal{B}_{\boldsymbol{2p}\cup\boldsymbol{3p}})$.
(iii) Optimize the neural negator (MLP_N) on queries involving negation (***2in***), with pretrained parameters frozen: $\mathcal{L}_3 = \mathcal{L}_{\text{CE}}(\mathcal{B}_{\boldsymbol{2in}}) + \mathcal{L}_{\text{NS}}(\mathcal{B}_{\boldsymbol{2in}}; \alpha) + \mathcal{L}_{\text{NL}}(\mathcal{B}_{\boldsymbol{2in}}; \beta)$.

5 Theoretical Analysis

Theorem 1 (Structural Expressivity for DAG-Structured EFO_1 Queries). *For any DAG-structured EFO_1 query under the query grammar considered in this work, LVSA can construct and persistently maintain a latent representation for every existential variable.*

Proof. We establish structural expressivity through constructive induction. Let $Q[V_?]$ be such a query with directed acyclic query graph $\mathcal{G}_Q = (\mathcal{V}_Q, \mathcal{E}_Q)$, where nodes $\mathcal{V}_Q$ represent anchor entities and variables, and edges $\mathcal{E}_Q$ encode relational literals. Following a topological ordering of $\mathcal{G}_Q$, LVSA recursively constructs latent states for all variables after their predecessor dependencies have been resolved.

Base Case: For anchor entities and independent existential variables, latent representations are directly computed by Eqs. (4) and (9), respectively.

Inductive Step: Assume that latent states are available for all predecessor nodes of a dependent existential variable $V_D \in \mathcal{V}_Q$. For the canonical dependency $F = \exists V_D : r_1(V', V_D) \wedge r_2(V_D, V)$, LVSA constructs the latent state of V_D via Eq. (12).

Canonicalization: Dependencies with multiple incident constraints can be organized through inverse-relation encoding and conjunction composition. Let V_D have forward dependencies F_D and backward dependencies B_D:

$$F_D = \exists V_D : \bigwedge_{i=1}^{N} r_{F,i}(u_i, V_D),\ B_D = \exists V_D : \bigwedge_{j=1}^{M} r_{B,j}(V_D, w_j). \tag{17}$$

Following the inverse-relation construction in Section 4.2, define $B_D^* = \exists V_D : \bigwedge_{m=1}^{M} r_{B,m}^{-1}(V_{I,m}, V_D)$. LVSA recursively aggregates these constraints:

$$\begin{aligned}\varphi(F_D) &= \varphi(r_{F,1} \otimes V_{F,1} + \cdots + r_{F,N} \otimes V_{F,N}), \\ \varphi(B_D^*) &= \varphi(r_{B,1}^{-1} \otimes V_{I,1} + \cdots + r_{B,M}^{-1} \otimes V_{I,M}), \\ \varphi(V_D) &= \text{MLP}_\text{D}\left(\varphi(F_D) \oplus \varphi(B_D^*)\right).\end{aligned} \tag{18}$$

LVSA combines conjunctive constraints through bundling, applies the neural negator to negated literals without altering the query-graph topology, and processes DNF disjuncts independently. Induction over a topological ordering of $\mathcal{G}_Q$ therefore shows that LVSA constructs and maintains a latent state for every existential variable in Q. □

Proposition 1 (Computational Complexity of Existential-State Construction). *Let $Q[V_?]$ involve k existential reasoning steps, and let d denote the embedding dimension. The computational complexity of LVSA's existential-state construction is $O(k \cdot d^2)$.*

Proof. At each existential reasoning step, LVSA performs: (i) relational projection via complex-valued binding operations, and (ii) construction of a persistent latent state for the existential variable through either MLP_I or MLP_D.

Relational projection requires $O(d)$ computation. Under fixed-depth MLPs with hidden size $O(d)$, both MLP_I and MLP_D require $O(d^2)$ computation. Hence, the dominant cost of each existential reasoning step is $O(d^2)$. For a query involving k existential reasoning steps, the total complexity is $O(k \cdot d^2)$. □

Complexity Discussion. LVSA applies a fixed-depth MLP at each existential reasoning step, yielding an existential-state construction cost of $O(k \cdot d^2)$ independent of the KG size $|\mathcal{V}|$. Scoring all candidate answers is a separate operation whose cost depends on $|\mathcal{V}|$. For comparison, GNN-based methods iteratively update query-graph states through message passing, Transformer-based methods apply attention over query components, and geometric methods rely on architecture-specific region transformations. Section 6.6 evaluates the end-to-end throughput of LVSA against representative GNN-based, geometric, and Grounding-based methods.

6 Experiments

6.1 Experimental Settings

Datasets. We evaluate LVSA on two standard CQA benchmarks: the suite introduced by Ren et al. [25] and the more recent benchmarks proposed by Gregucci et al. [10]. These benchmarks are constructed from FB15k [5], FB15k-237 [30], and NELL995 [35] using their official splits. Detailed statistics are provided in Table 1. The standard benchmarks primarily contain partially observable queries, whereas the newer benchmarks additionally introduce fully unobservable queries that demand stronger compositional generalization. In total, the evaluation covers 14 basic query types with existential quantification and logical negation; the newer benchmarks further add more complex structures (***4p*** and ***4i***). Comprehensive query descriptions are shown in Figure 3.

Evaluation Protocol. Following Ren et al. [25], we adopt filtered Mean Reciprocal Rank (MRR) and Hits@K (H@K) as evaluation

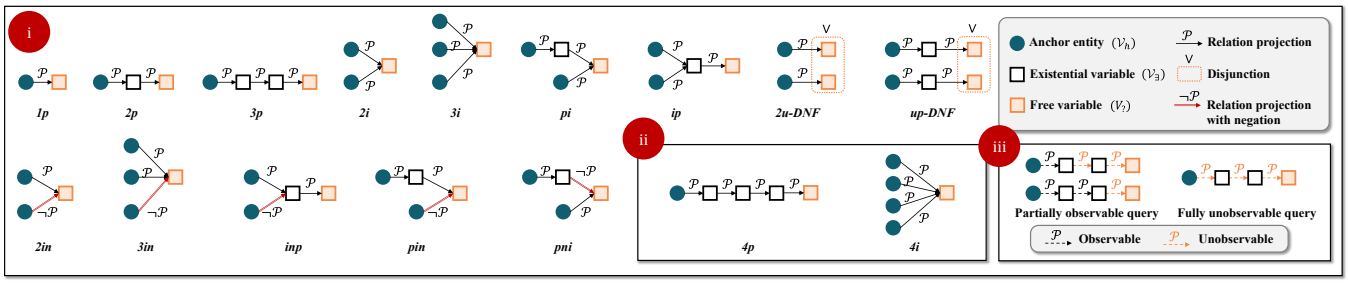


Figure 3: (i) Query graphs of 14 standard query types. Specifically, query types labeled with "*p*" (except the atomic query *1p*) contain existential quantifiers. Besides, "*i*", "*u*", and "*n*" indicate the presence of logical conjunction, disjunction, and negation operations, respectively. (ii) Query graphs of novel query types from new benchmarks. (iii) Samples of partially observable queries in the standard benchmarks and fully unobservable queries in the new benchmarks.

Table 1: Statistics of the adopted datasets. - indicates that the corresponding dataset is not provided in the benchmark.

Standard Datasets [24]	#Entity	#Relation	#Training Query		#Validation Query		#Testing Query		New Datasets [10]	#Testing Query in Total		
			1p/2p/3p/2in	Others	*1p*	Others	*1p*	Others		14 Types	*4p*	*4i*
FB15k	14,951	1,345	273,710	273,71	59,097	8,000	67,016	8,000	-	-	-	-
FB15k-237	14,505	237	149,689	149,68	20,101	5,000	22,812	5,000	**FB15k-237+H**	68,687	2,670	20,189
NELL995	63,361	200	107,982	107,98	16,927	4,000	17,034	4,000	**NELL995+H**	71,464	2,322	22,326

metrics. For a query $Q[V_?]$ with answer set $A[Q] \subseteq \mathcal{V}$:

$$\text{MRR} = \frac{1}{|A[Q]|} \sum_{a \in A[Q]} \frac{1}{\text{Rank}(a \mid \mathcal{V} \setminus A[Q])}, \tag{19}$$

$$\text{H@}K = \frac{1}{|A[Q]|} \sum_{a \in A[Q]} \mathbf{1}\left[\text{Rank}(a \mid \mathcal{V} \setminus A[Q]) \leq K\right], \tag{20}$$

where $\text{Rank}(a \mid \mathcal{V} \setminus A[Q])$ denotes the filtered ranking position of answer entity a among all candidate entities after removing other correct answers in $A[Q]$.

Baselines. We compare LVSA against representative Skolemization-based methods (CQD-CO [1], BetaE [25], ConE [38], LMPNN [33], and CLMPT [37])[2] as well as Grounding-based methods (CQD-Beam [1], GNN-QE [40], ULTRAQ [9], and QTO [3]). In line with prior work, most baselines employ ComplEx embeddings [31], with the exception of BetaE and ConE.

Implementation. All experiments are conducted on a single Nvidia RTX 3090 GPU with 24GB memory. LVSA applies LeakyReLU activations for all neural modules. Based on a grid search, we set the regularization coefficients to $\alpha = \beta = 1$ and optimize all stages using Adam with a learning rate of 5×10^{-4}.

As described in Section 4.1, LVSA's embedding space and triple scoring metric are aligned with ComplEx. This enables us to initialize LVSA's entity and relation embeddings using pre-trained ComplEx embeddings ($d = 1000$). In subsequent stages, optimization is restricted to the differentiable Skolemization module and the neural negator. To further stabilize the training of the differentiable Skolemization module, we introduce an auxiliary cross-entropy objective over in-batch candidate entities $\mathcal{V}_\mathcal{B} = \{a \in A[Q]\}_{(Q,a)\in\mathcal{B}}$:

$$\mathcal{L}_{\text{CE}}(\mathcal{B}_{2p\cup 3p}) = \frac{1}{|\mathcal{B}|} \sum_{Q \in \mathcal{B}} \Big(H(Q, \mathcal{V}) + H(Q, \mathcal{V}_\mathcal{B}) \Big). \tag{21}$$

[2]We additionally consider the recent Transformer-based method PPMT [41]. Since its original experiments adopt a different data split and involve substantially longer training time (>48h per dataset), we provide supplementary reproduction results in our github repository.

6.2 Main Results

Tables 2 and 3 present the main results of the proposed LVSA against Skolemization-based baselines on standard benchmarks.

Comparison with Prior Skolemization-based Methods. Among existing Skolemization-based approaches, CQD-CO [1] and geometric embedding methods such as BetaE [25] and ConE [38] achieve weaker performance, especially on queries involving logical negation. CQD-CO relies on simplified latent existential optimization, while geometric embedding methods may suffer from misalignment between geometric inductive biases and compositional logical semantics. GNN-based methods, including LMPNN [33] and CLMPT [37], achieve competitive performance on several query types, although their results vary substantially across datasets.

Overall Superiority of LVSA. Across all three datasets, LVSA consistently achieves the best overall performance, with statistically significant improvements over all baselines (paired t-test, $p < 0.01$). The improvements are particularly pronounced on conjunction-heavy queries such as ***2i*** and negation queries such as ***2in***. These results demonstrate the effectiveness of LVSA's binding and bundling operations and its logic-regularized neural negator for modeling negation. LVSA also achieves competitive performance on multi-hop queries while maintaining explicit existential assignments throughout reasoning. A detailed evaluation of existential consistency is provided in Section 6.4.

Table 2: The MRR (%) results of our LVSA and the existing Skolemization-based baselines on the standard benchmarks. A_p, A_n, and A are the average scores among EPFO queries, queries involving negation, and all queries, respectively.

Datasets	Model	A_p	A_n	A	*1p*	*2p*	*3p*	*2i*	*3i*	*pi*	*ip*	*2u*	*up*	*2in*	*3in*	*inp*	*pin*	*pni*
FB15k	CQD-CO	45.3	4.6	30.8	**89.4**	27.6	15.1	63.0	65.5	46.0	35.2	42.9	23.2	0.2	0.2	4.0	0.1	18.4
	BetaE	41.6	11.8	31.0	65.1	25.7	24.7	55.8	66.5	43.9	28.1	40.1	25.2	14.3	14.7	11.5	6.5	12.4
	ConE	49.8	14.8	37.3	73.3	33.8	29.2	64.4	73.7	50.9	35.7	55.7	31.4	17.9	18.7	12.5	9.8	15.1
	LMPNN	50.6	20.0	39.7	85.0	39.3	28.6	68.2	76.5	46.7	43.0	36.7	31.4	29.1	29.4	14.9	10.2	16.4
	CLMPT	55.1	17.9	41.8	86.1	**43.3**	**33.9**	69.1	78.2	55.1	**46.6**	46.1	**37.3**	21.2	22.9	**17.0**	13.0	15.3
	LVSA	**59.9**	**27.7**	**48.4**	**89.4**	40.6	32.0	**80.4**	**84.2**	**61.1**	43.2	**72.0**	35.8	**41.5**	**33.7**	15.0	**17.9**	**30.4**
FB15k-237	CQD-CO	19.8	1.7	13.3	46.7	10.3	6.5	23.1	29.8	22.1	16.3	14.2	8.9	0.2	0.2	2.1	0.1	6.1
	BetaE	20.9	5.5	15.4	39.0	10.9	10.0	28.8	42.5	22.4	12.6	12.4	9.7	5.1	7.9	7.4	3.5	3.4
	ConE	23.4	5.9	17.1	41.8	12.8	11.0	32.6	47.3	25.5	14.0	14.5	10.8	5.4	8.6	7.8	4.0	3.6
	LMPNN	24.1	7.8	18.3	45.9	13.1	10.3	34.8	48.9	22.7	17.6	13.5	10.3	8.7	12.9	7.7	4.6	5.2
	CLMPT	25.9	7.9	19.4	45.7	13.7	11.3	37.4	52.0	28.2	**19.0**	14.3	11.1	7.7	13.7	8.0	5.0	5.1
	LVSA	**28.8**	**10.9**	**22.4**	**49.2**	**15.8**	**12.1**	**41.7**	**55.1**	**31.0**	**19.0**	**22.5**	**12.4**	**12.4**	**18.3**	**8.5**	**7.3**	**8.0**
NELL995	CQD-CO	28.4	1.9	18.9	60.8	18.3	13.2	36.5	43.0	30.0	22.5	17.6	13.7	0.1	0.1	4.0	0.0	5.2
	BetaE	24.6	5.9	17.9	53.0	13.0	11.4	37.6	47.5	24.1	14.3	12.2	8.5	5.1	7.8	10.0	3.1	3.5
	ConE	27.1	6.4	19.7	53.1	16.1	13.9	40.0	50.8	26.3	17.5	15.3	11.3	5.7	8.1	10.8	3.5	3.9
	LMPNN	30.7	8.0	22.6	60.6	22.1	17.5	40.1	50.3	28.4	24.9	17.2	15.7	8.5	10.8	**12.2**	3.9	4.8
	CLMPT	31.3	7.0	22.6	58.9	22.1	18.4	41.8	51.9	28.8	24.4	18.6	16.2	6.6	8.1	11.8	3.8	4.5
	LVSA	**32.4**	**9.1**	**24.1**	**61.0**	**22.2**	**18.9**	**43.6**	**53.2**	**31.8**	**25.0**	**19.4**	**16.6**	**10.1**	**11.4**	11.8	**6.1**	**6.1**

Table 3: H@1 (%) of LVSA and Skolemization-based baselines on standard benchmarks.

Model	FB15k			FB15k-237			NELL995		
	A_p	A_n	A	A_p	A_n	A	A_p	A_n	A
CQD-CO	39.7	1.9	26.2	14.7	0.5	9.7	21.3	0.7	14.0
BetaE	31.3	5.2	21.9	13.4	2.8	9.6	17.8	2.1	12.2
ConE	39.6	7.3	28.0	15.6	2.2	10.9	19.8	2.2	13.5
LMPNN	41.4	12.2	31.0	16.3	3.5	11.7	23.0	3.4	16.0
CLMPT	45.9	9.3	32.9	17.4	2.9	12.2	22.7	2.4	15.4
LVSA	**53.2**	**18.9**	**41.0**	**20.8**	**5.5**	**15.4**	**24.3**	**3.8**	**17.0**

Failure-Case Analysis. We examine the query types on which LVSA does not lead or achieves its smallest gains. Although LVSA slightly underperforms CLMPT on FB15k's ***2p*** and ***3p*** queries, this may be related to inverse-relation leakage in FB15k [30], where answers can often be inferred via relation patterns without requiring full multi-hop composition. In contrast, LVSA's explicit maintenance of persistent latent assignments reduces reliance on such shortcuts, leading to more stable performance across different datasets. LVSA also exhibits relatively smaller gains on ***ip*** and ***inp*** queries, where conjunction-composed representations are directly fed into differentiable Skolemization. These results identify complex compositional inputs to the differentiable Skolemization module as a recurring source of smaller gains.

6.3 Results on New Benchmarks

The new benchmarks extend the standard evaluation on partially observable queries (Figure 3(i)) by targeting fully unobservable queries (Figure 3(iii)) and introducing more complex query types (Figure 3(ii)). We evaluate LVSA against the strongest Grounding-based and Skolemization-based methods to further assess generalization under challenging reasoning settings.

Evaluation on Fully Unobservable Queries. Table 4 reveals a notable shift in the relative performance of the two reasoning paradigms on fully unobservable queries. Although Skolemization-based methods achieve slightly lower performance than Grounding-based methods under partial observability (A: 19.6% vs. 22.1% on FB15k-237), they obtain higher MRR on fully unobservable queries (A_H: 10.7% vs. 9.0% on FB15k-237+H) and exhibit substantially smaller performance degradation when transferring to the new benchmarks ($\triangle$: 46.0% vs. 58.6%).

Grounding-based methods rely on explicit entity-level matching, which benefits from observable evidence but may be limited under missing or incomplete observations. In contrast, Skolemization-based methods perform latent compositional reasoning, which may contribute to their smaller performance degradation under reduced observability on these benchmarks. Among all evaluated models, LVSA achieves the best overall performance on fully unobservable queries, suggesting improved generalization under knowledge-sparse settings.

Evaluation on More Complex Query Structures. Consistent with the results on the 14 standard query types, Skolemization-based methods outperform Grounding-based approaches on the more complex ***4i*** and ***4p*** queries from the new benchmarks, with LVSA achieving the best overall performance (Figure 4). These results further validate the effectiveness of LVSA's differentiable Skolemization module and bundling mechanism.

6.4 Evaluation of Existential Consistency

Table 4: The MRR (%) results on new benchmarks, with relative decline (△) from standard benchmarks. $A_{p,H}$, $A_{n,H}$, and A_H represent the average scores across EPFO, negation, and all 14 standard query types on the new benchmarks, respectively.

Paradigm	Model	FB15k-237+H					NELL995+H				
		$A_{p,H}$	$A_{n,H}$	A_H	A	△(↓)	$A_{p,H}$	$A_{n,H}$	A_H	A	△(↓)
Grounding -based	GNN-QE	12.6	3.1	9.2	20.9	55.9	16.9	2.7	11.9	22.1	46.3
	ULTRAQ	11.4	2.4	8.2	18.4	**55.5**	11.8	2.3	8.4	17.1	50.9
	QTO	**12.7**	**4.2**	**9.7**	**27.1**	64.3	**20.2**	**5.0**	**14.8**	**25.7**	**42.6**
	Avg.	*12.2*	*3.2*	*9.0*	*22.1*	*58.6*	*16.3*	*3.3*	*11.7*	*21.6*	*46.6*
Skolemization -based	ConE	11.6	2.8	8.5	17.1	50.5	19.6	3.2	13.8	19.7	30.2
	CLMPT	15.6	3.5	11.3	19.4	**41.9**	24.1	3.4	16.7	22.6	25.9
	LVSA	**15.7**	**5.8**	**12.2**	**22.4**	45.6	**25.5**	**5.0**	**18.2**	**24.1**	**24.6**
	Avg.	*14.3*	*4.0*	*10.7*	*19.6*	*46.0*	*23.1*	*3.9*	*16.2*	*22.1*	*26.9*

* The performance decline △% is defined as $(A - A_H)/A \times 100\%$.

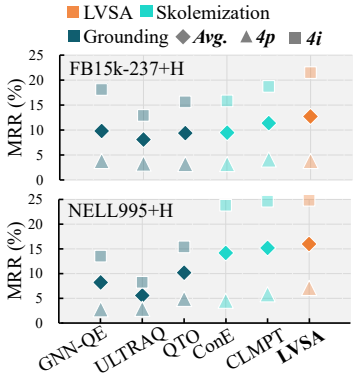


Figure 4: The MRR (%) results of *4p* and *4i* queries.

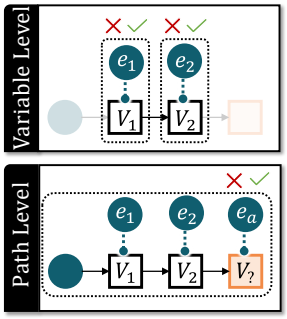


Figure 5: Two-level existential consistency evaluation.

Table 5: Existential consistency evaluation (%) on multi-hop queries at variable and path levels.

Datasets	Model	Variable-level MRR				Variable-level P@1				Path-level P@1		
		2p	*3p*		*Avg.*	*2p*	*3p*		*Avg.*	*2p*	*3p*	*Avg.*
		V_1	V_1	V_2		V_1	V_1	V_2				
FB15k	LVSA	**57.8**	60.2	**40.1**	**52.7**	**50.9**	59.4	**33.2**	**47.8**	**69.9**	**40.7**	**55.3**
	CLMPT	53.3	**63.5**	35.3	50.7	45.9	**61.2**	28.1	45.1	62.4	38.0	50.2
	ConE	44.2	47.3	24.2	35.8	35.6	43.0	17.4	30.2	0.8	0.1	0.5
FB15k-237	LVSA	**47.6**	**52.4**	**31.7**	**43.9**	**40.3**	**51.3**	**24.9**	**38.8**	**66.1**	**34.4**	**50.3**
	CLMPT	40.8	49.3	23.8	38.0	31.9	46.9	16.7	31.8	41.2	19.0	30.1
	ConE	34.1	38.9	16.2	27.6	25.2	33.1	10.2	21.7	0.7	0.1	0.4
NELL995	LVSA	**68.3**	61.7	**47.1**	**59.0**	**63.4**	**61.9**	**41.2**	**55.5**	**83.4**	**57.6**	**70.5**
	CLMPT	63.8	**62.3**	45.1	57.1	56.8	61.7	35.6	51.4	66.2	38.9	52.6
	ConE	50.4	42.0	19.6	30.8	42.8	37.5	12.3	24.9	1.1	0.2	0.7

We compare LVSA against the best-performing geometric embedding-based method (ConE) and GNN-based method (CLMPT) to evaluate *empirical existential consistency* throughout multi-step reasoning. We operationalize it at two levels: variable-level metrics examine whether each latent state can be decoded to a valid intermediate entity, while path-level metrics examine whether the jointly decoded assignments form a valid reasoning path.

Quantitative Evaluation. We propose a two-level evaluation framework (Figure 5). At the *variable level*, we measure whether each existential variable is instantiated to a valid intermediate entity using MRR and P@1. At the *path level*, we assess whether the entire instantiated reasoning chain is valid using P@1.

As shown in Table 5, LVSA outperforms CLMPT and ConE at both levels across all datasets, providing evidence of stronger empirical existential consistency throughout multi-hop reasoning. While CLMPT matches LVSA when instantiating the first variable V_1 in ***3p*** queries, its performance drops substantially on V_2 and complete reasoning paths. ConE exhibits an even sharper degradation trend. These results suggest that prior Skolemization-based methods progressively entangle existential information within query-level representations, making it difficult to maintain coherent latent assignments over long compositional reasoning chains. In contrast, LVSA preserves persistent per-variable latent assignments, resulting in substantially stronger path-level reasoning accuracy.

Qualitative Evaluation. Figure 6 presents a case study comparing how LVSA and CLMPT instantiate existential variables via $V \approx \arg\max_{e \in \mathcal{V}} \phi_E(\varphi(V), \varphi(e))$. LVSA correctly resolves the intermediate entity as Computer, while CLMPT incorrectly associates it with a music album. Although both models reach the correct final answer (Cisco), CLMPT's incorrect intermediate assignment reveals a reasoning shortcut.

6.5 Ablation Analysis

VSA-based composition and differentiable Skolemization jointly form LVSA's core reasoning architecture. We therefore focus the ablation study on the two logic-guided regularization objectives of the neural negator, which can be removed independently while leaving the core architecture unchanged.

As shown in Table 6, removing the satisfiability loss $\mathcal{L}_{\text{NS}}$ leads to a slight performance drop, indicating its contribution to preserving negation satisfiability. In contrast, removing the logical-axiom loss $\mathcal{L}_{\text{NL}}$, which encourages the double-negation and contradiction laws, causes the largest performance degradation. Specifically, the average MRR on negation queries decreases by 5.0%, 3.3%, and 2.1%

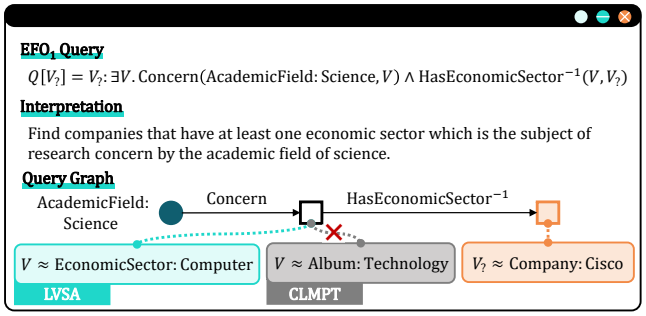


Figure 6: Instantiating latent assignments of existential variables: LVSA vs. CLMPT.

Table 6: The MRR (%) results of our LVSA and ablation models across three standard datasets.

Datasets	Model	A_n	*2in*	*3in*	*inp*	*pin*	*pni*
FB15k	LVSA	27.7	41.5	33.7	15.0	17.9	30.4
	w/o $\mathcal{L}_{\text{NS}}$	25.6	39.0	29.9	14.0	16.5	28.7
	w/o $\mathcal{L}_{\text{NL}}$	22.7	24.6	30.1	12.5	13.1	19.0
FB15k -237	LVSA	10.9	12.4	18.3	8.5	7.3	8.0
	w/o $\mathcal{L}_{\text{NS}}$	10.4	12.2	16.5	8.1	6.9	8.4
	w/o $\mathcal{L}_{\text{NL}}$	7.6	7.4	13.9	7.0	5.1	4.5
NELL995	LVSA	9.1	10.1	11.4	11.8	6.1	6.1
	w/o $\mathcal{L}_{\text{NS}}$	9.0	9.9	11.1	11.8	6.1	6.2
	w/o $\mathcal{L}_{\text{NL}}$	7.0	6.6	10.8	9.4	3.9	4.4

on FB15k, FB15k-237, and NELL995, respectively. Consistent trends are observed on the two new benchmarks. Overall, these results demonstrate the complementary contributions of satisfiability and logical-axiom regularization to neural negation.

6.6 Efficiency Analysis

We complement Proposition 1 with an empirical efficiency comparison against CLMPT, ConE, and CQD-Beam [1].

Figure 7 compares average MRR and Queries Per Second (QPS) on existential positive queries from FB15k-237. The comparison highlights distinct performance–efficiency tradeoffs among geometric, GNN-based, and Grounding-based architectures. LVSA outperforms CLMPT while achieving 2.6× and 1.7× the throughput of CLMPT and ConE, respectively. This advantage is consistent with LVSA's lightweight algebraic computation, compared with the iterative GNN–Transformer updates in CLMPT and the geometric transformations in ConE. For CQD-Beam, smaller beams improve efficiency at the cost of completeness and performance. Its throughput falls below all evaluated Skolemization-based methods when $b \geq 128$. For $64 \leq b \leq 256$, LVSA achieves comparable MRR with 2–15× higher QPS.

Figure 8 further examines scaling from ***2p*** to ***3p***. LVSA and ConE show marginal QPS reductions, whereas CLMPT drops by 55.4%. CQD-Beam's fitted QPS declines nearly ten times faster on ***3p*** than on ***2p*** as beam size grows, and it encounters Out-Of-Memory errors when $b \geq 512$ and $k > 2$. Together, these results empirically

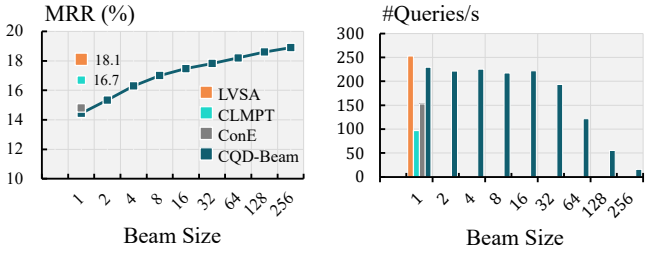


Figure 7: The performance (left) and throughput (right) of LVSA, CLMPT, ConE, and CQD-Beam on existential positive queries (*2p/3p/ip/pi/up*) in FB15k-237.

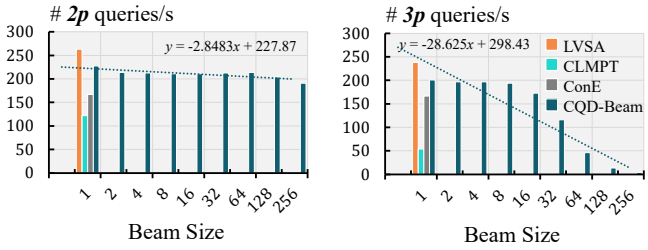


Figure 8: The throughput of LVSA and baseline models on *2p* (left) and *3p* (right) queries in FB15k-237. The dashed line shows CQD-Beam's performance decline as beam size grows.

support the efficiency advantage of avoiding entity enumeration during existential-state construction.

7 Conclusion

We introduce the Grounding–Skolemization dichotomy to characterize the fundamental tradeoff between logical fidelity and computational efficiency in CQA over incomplete KGs. Building upon this perspective, we propose LVSA, a neuro-symbolic framework for existentially consistent Skolemized reasoning. The theoretical analysis establishes LVSA's structural expressivity for DAG-structured EFO_1 queries, while experiments demonstrate strong reasoning performance, efficiency, and empirical existential consistency. Current limitations include the robustness of differentiable Skolemization under highly compositional inputs and its ability to represent diverse valid witnesses in multi-answer queries. Future work will explore more robust state construction and richer representations of multiple witnesses. We hope this work provides a foundation for scalable and logically faithful latent reasoning over incomplete KGs.

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Generative AI tools were used solely to improve the language and presentation of this manuscript. All AI-assisted revisions were reviewed and approved by the authors, who take full responsibility for the content of the paper.

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